جـامـعـة الـشـرق الأوسـط
MIDDLE EAST UNIVERSITY
Amman - Jordan

Course Syllabus

General Information

Faculty	Information Technology		
Department	Cybersecurity Dept.		
Semester	2nd	**Academic Year**	2025/2026

Course Details

Course Title	Application Security			**Course No.**	0454302		
Type of Learning	☐ On- campus ■ Blended (................ Platform) ☐ Online (............... Platform)						
Credit Hours	3	**Theoretical**	1	**Practical**	2		
Course Level (according to JNQF)			6	**JNQF Hours***	12.15		
Pre-requisite	0413201- Web Applications			**Co-requisite**			

Course Instructor Information

Instructor Name	Madallah Al-Majali		
Instructor E-mail	malmajali@meu.edu.jo		
Course Time	08:00-10:00 (Sun, Tues) 08:00-10:00 (Mon, Wed)		
Office Hours	8-10 (Thr)		
Office Number	N346	**Office Phone**	315
Lab Instructor Name –If assigned-	--		

Sources and References

1. Textbook

Tanya Janca, 2025 , Alice & Bob Learn Application Security , 1'st Edition , Publisher : Wiley; ISBN-10 : 1119687357

2. References

1-R. Sarma Danturthi (2024) , Database and Application Security: A Practitioner's Guide, Publisher : Addison-Wesley Professional; 1st edition, ISBN-13 : 978-0138073732

2- Andrew Hoffman (2020), Web Application Security Exploitation and Countermeasures for Modern Web Applications , **Publisher :** O'Reilly Media , ISBN-13 : 978-1492053118

Teaching Methods

1. Lectures	**2.** Questions	**3**. Reports	**4**. Exercises

Course Description

Explore the essential principles of building secure software by following the journey of Alice and Bob as they navigate the complex world of application security. Drawing from real-world scenarios and guided by industry best practices, this course introduces students to secure software development life cycles (SSDLC), threat modeling, secure design, and common vulnerabilities such as injection, cross-site scripting (XSS), broken authentication, and more. Students will gain hands-on experience identifying, mitigating, and preventing security flaws in modern web applications. Emphasis is placed on integrating security into every phase of development, empowering developers to create resilient, secure, and trustworthy systems.

Course Learning Outcomes (CLOs) and its correlation with Program Learning Outcomes (PLOs) and JNQF Descriptors

No.	CLO's	PLO's	JNQF Descriptors		
			Knowledge	Skill	Competency
1-	**Identify security principles** such as least privilege, secure defaults, defense in depth, and fail securely during the development and deployment of a secure web system.	6		S2-I	
2-	**Demonstrate the ability to maintain operations** in a secure web application by applying runtime protections, monitoring strategies, and secure configuration practices in the presence of evolving threats.	6		S2-D	
3-	**Design security features** for a web application by analyzing common threats and aligning solutions with secure development best practices.	2		S1-D	
4-	**Integrate security controls** in a web application to address vulnerabilities such as injection attacks, cross-site scripting (XSS), broken authentication, and insecure data storage.	2		S1-P	
5-	**Identify the security posture** of a web application using manual and automated testing tools to identify and mitigate potential risks.	2		S1-I	

Course Topics

Week	Topics	Lecture Type		CLOs	Assessment	
		Lecture No.	Lecture Type		Assessment Tool	Summative/ Formative
1	Introduction to course and syllabus	1-3	Lecture	1	Discussion & Questions	Formative
2	-Introduction to Application Security -Introduction to SSDLC	4-6	Lecture	1	Discussion & Questions	Formative
3	-Security Requirements Gathering -Identifying Security Requirements	7-10	Lecture	1	Discussion & Questions	Formative
4	Least Privilege and Defense in Depth Security Requirements	11-13	Lecture	1	Discussion & Questions Quiz	Formative Summative
5	OWASP	14-16	Lecture	2,3	Discussion & Questions	Formative
6	ASVS OWASP OWASP OTG Guide	17-19	Lecture	2,3	Discussion & Questions	Formative
7	-Secure Application Design and Architecture -Secure Architecture Patterns	20-22	Lecture	2,3	Discussion & Questions Assignments	Formative Summative
8	**Mid-Exam**	23-24	Assessment	1,2,3	Questions	Summative
9	Secure Coding Practices for Input Validation Preventing Injection Attacks	25-27	Lecture	3,4	Discussion & Questions	Formative
10	Secure Coding Practices for Authentication and Authorization	28-30	Lecture	3,4	Discussion & Questions Quiz	Formative Summative
11	-Role-Based Access Control (RBAC) -Secure Coding Practices for Cryptography	31-33	Lecture	3,4	Discussion & Questions	Formative
12	Secure Coding Practices for Session Management Secure Session Cookies	34-36	Lecture	3,4	Discussion & Questions Assignments	Formative Summative
13	Static and Dynamic Application Security Testing (SAST & DAST)	37-39	Lecture	3,4	Discussion & Questions	Formative
14	Secure Deployment and Maintenance	40-42	Lecture	3,4	Discussion & Questions	Formative
15	**Project Presentation**	46-48	Lecture	5	Discussion & Questions	Summative
16	**Final Exam**	-	**Assessment**	**1,2,3,4,5**	Questions	**Summative**

Assessment Tool	**Weight %**	**Expected Due Date**
Mid Exam	25%	Week 8

Quiz	20%	Week 4, 10
Project	15%	Week 15
Final Exam	40%	Week 16

Course Policies

Item	Policy
Quizzes	▪ No makeup quizzes. ▪ A minimum 2 quizzes are given. ▪ Each Quiz is out of 5. ▪ If five quizzes or more are given then the lowest quiz's grade is dropped.

Exams	▪ The format of the exams is generally (but NOT always) as follows: General Definitions, Multiple-Choice, True/False, Analyze a Problem, Essay Questions, etc.
Makeup Exams	▪ Makeup exam should not be given unless there is a valid excuse.
Drop Date	▪ Last day to drop the course is according to the University regulations.
Academic Honesty	▪ Cheating or copying on exam or quiz is an illegal and unethical activity. ▪ Standard MEU policy will be applied. ▪ All graded assignments must be your own work (your own words).
Attendance	▪ Excellent attendance is expected. ▪ MEU policy requires the faculty member to assign ZERO grade if a student misses 20% of the classes. ▪ Sign-in sheets will be circulated. ▪ If you miss class, it is your responsibility to find out about any announcements or assignments you may have missed.
E-Learning	▪ It is your responsibility to check the course's eLearning website regularly (at least once every 2-3 days) for announcements and material.
Workload	▪ The student should expect to spend 6 hours per week as an average workload.
Graded Exams	▪ Instructor should return exam papers graded to students within one week after the exam date.
Participation	▪ Participation in and contribution to class discussions will affect your final grade positively. Raise your hand if you have any questions. ▪ Making any kind of disruption and (side talks) in the class will affect you negatively.
Assignment/ Projects	▪ Assignments are as mentioned in the previous table. ▪ Students organize themselves in teams of (2-3) students each. Each team will give at least one presentation regarding a selected case, delivering a fully working application. ▪ More details about phases and topics of the projects will be given later on.
Others	

Rubrics and Feedbacks (If Applicable)

Answers assessments will be provided after assessments and during the lecture.

Reports and Presentation

Criteria	Excellent (13-15 points)	Good (10-12 points)	Satisfactory (7-9 points)	Needs Improvement (0-6 points)
		Report (60%)		
Introduction (10%)	Clear, concise introduction with a well-defined problem statement and objectives.	The introduction is clear but lacks some detail or clarity in the problem statement and objectives.	The introduction is present but lacks clarity and detail.	The introduction is unclear or missing.
Design Alternatives and Constraints (15%)	Comprehensive analysis of design alternatives with clear identification of constraints.	Good analysis of design alternatives but lacks some detail or clarity in constraints.	Fundamental analysis of design alternatives with minimal identification of constraints.	Analysis of design alternatives and constraints is unclear or missing.
Methodology (15%)	Detailed and well-structured methodology with clear steps and justification.	Clear methodology but lacks some detail or justification.	Basic methodology with minimal detail.	The methodology is unclear or missing.
Results and Analysis (15%)	Clear presentation of results with thorough analysis and interpretation.	There are precise results, but the analysis lacks depth.	Fundamental presentation of results with minimal analysis.	Results are unclear, or analysis is missing.
Conclusion and Recommendations (10%)	Clear and concise conclusion with well-supported recommendations.	Clear conclusion, but the recommendations lack some support.	Basic conclusion with minimal recommendations.	The conclusion is unclear or missing.
References and Appendices (5%)	All references and appendices are complete and correctly formatted.	Minor errors in references or appendices.	Some references or appendices are missing or incorrectly formatted.	References and appendices are incomplete or missing.

Criteria	Excellent (9-10 points)	Good (7-8 points)	Satisfactory (5-6 points)	Needs Improvement (0-4 points)
		Presentation (40%)		
Content (15%)	Comprehensive and well-organized content with clear vital points.	Clear content but lacks some organization or critical points.	Primary content with minimal organization.	Content is unclear or disorganized.
Delivery (10%)	Confident and engaging delivery with good eye contact and clear speech.	Clear delivery but lacks some confidence or engagement.	Essential delivery with minimal engagement.	Delivery is unclear or disengaging.
Visual Aids (10%)	Practical and well-designed visual aids that enhance the presentation.	Good visual aids but lack some design or effectiveness.	Basic visual aids with minimal design.	Visual aids are unclear or ineffective.
Q&A (5%)	Responds to questions confidently and accurately.	Responds to questions but lacks some confidence or accuracy.	Primary responses to questions with minimal detail.	Responses are unclear or inaccurate.

*** JNQF Hours Calculation**

Activity	Duration/ week	Times/ Semester	No. of Hours
Learning Activities			
Lectures and Seminars	3 hours	15	45
Tutorials	0	0	0
laboratory	0	0	0
Activities for Assessment and Evaluation			
First Exam/ Midterm	1.5 hours	1	1.5
Second Exam (If applicable)	0	0	0
Final Exam	2 hours	1	2
Quizzes	15 minutes	2	0.5
Self-Learning Activities			
Assignments/ Homework	2.5 hours	2	5
Case Studies	0	0	0
Presentations	0	0	0
E-learning	0	0	0
Extra Readings	4.5	15	67.5
Total Hours	121.5		
JNQF Hours	=121.5/10 = 12.15		